



ABDULLAH RACHMAT SUGI

GEOMATICS ENGINEERING | GIS | SURVEY & MAPPING

Surabaya, East Java, Indonesia

+62 858 5359 9295 | rachmatsugi09@gmail.com

Portfolio: its.id/m/rachmatsugi

SUMMARY

Fresh Graduate in Geomatics Engineering from Institut Teknologi Sepuluh Nopember (ITS) with a GPA of 3.41/4.00 and a Very Satisfactory distinction, with experience in GIS, surveying and mapping, and geospatial data management and analysis. Experienced through internships at ATR/BPN and PT Kereta Api Indonesia in GNSS surveying, land asset mapping and verification, cadastral data validation, and spatial data processing using ArcGIS and AutoCAD. Additional project experience includes remote sensing, UAV photogrammetry, and 3D modeling, with the ability to process, analyze, and validate geospatial data to produce accurate spatial information and support mapping and analytical needs.

EDUCATION

Sepuluh Nopember Institute of Technology

Jul 2022 - Aug 2026

Bachelor of Engineering in Geomatics Engineering

- GPA: 3.41/4.00
- Thesis on "3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method"

SMA Negeri 1 Bangsal

Jul 2019 - May 2026

Science Major (MIPA)

- Average Grade : 90.23/100

INTERNSHIP EXPERIENCE

PT Kereta Api Indonesia (Persero) Regional Operation VII Madiun

Jul 2025 - Aug 2025

- Conducted terrestrial boundary surveys for PT KAI land assets on approximately 30 parcels in the Jombang Station area and Jl. Anggrek Madiun using EFIX GNSS RTK for field boundary verification.
- Processed survey data and spatial digitization using AutoCAD Civil 3D and ArcGIS Pro to delineate asset boundaries, integrate administrative data, and analyze corridor areas for operational and non-operational lines.
- Developed 6 thematic maps and 1 survey layout depicting station distributions, land asset boundaries, and railway corridors across Daop 7 Madiun.

National Land Agency (ATR/BPN) of Kediri City

Jan 2025 - Feb 2025

- Processed and updated cadastral maps using AutoCAD and ArcGIS to support land registration data maintenance across Kediri City.
- Conducted terrestrial field boundary surveys using GNSS for residential housing areas and built structures in Kediri City for on-site boundary verification.
- Cross-checked spatial parcel data against land certificates (SHM) to ensure consistency between physical boundaries and legal ownership documents.

National Land Agency (ATR/BPN) of Batu City

Jul 2024 - Aug 2024

- Performed topological validation and data cleaning on 115 land parcels to eliminate gaps and overlaps, ensuring spatial accuracy and database consistency.
- Conducted field boundary surveys using geodetic GNSS equipment across 10 plantation parcels in Sumberbrantas Village, Bumiaji District.
- Validated land ownership records by cross-referencing freehold certificates (Sertifikat Hak Milik/SHM) with the institutional cadastral database to ensure legal tenure and data integrity.

ORGANIZATIONAL EXPERIENCE

Student Legislative Board, Faculty of Civil, Planning, and Geo Engineering (BLM FT-SPK ITS)	Nov 2024 - Feb 2026
- Responsible for developing the Organizational Work Guidelines (GBHK) for HIMAGE-ITS and providing guidance to commission members. - Conducted policy and regulatory reviews, drafting, and evaluation to support effective organizational governance.	
Student Representative Council, Geomatics Engineering Student Association (BPM HIMAGE-ITS)	Dec 2023 - Dec 2025
- Served as Session Chair II and contributed to the implementation of the High-Level Deliberation (MTT) of FT-SPK ITS. - Managed official correspondence and administrative processes while establishing strategic communication with organizational stakeholders.	

CERTIFICATIONS & TRAINING

Small Unmanned Aircraft System Remote Pilot Course
PT Terra Drone Indonesia 2026
Drone Basic Knowledge & Drone Training Mapping
PVIA ITS 2023–2024
Satellite Imagery Data Processing with ENVI
HIMAGE-ITS 2024

PROJECTS

3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method	Nov 2025 - June 2026
Bachelor's Thesis Project	
- Integrated 1,543 UAV and terrestrial photogrammetric images using 3D Gaussian Splatting (3DGS) with 20,000 optimization iterations. - Produced a model 70% smaller with 4–6× faster rendering compared to SfM/MVS. - Achieved 1.36 cm RMSE and 0.183 m Cloud-to-Cloud deviation, validated across 7 field measurement segments.	
Spatio-Temporal Monitoring of Wonorejo Aquaculture Ponds	Feb 2025 - May 2025
2025 Thematic Fieldwork – Department of Geomatics Engineering, ITS	
- Processed multitemporal Sentinel-2A imagery (2015–2024) using SVM in ENVI for land-cover classification. - Executed confusion matrix-based accuracy assessment, achieving an Overall Accuracy of up to 81.26% and a Kappa coefficient of 0.66. - Produced cartographically standardized multitemporal thematic maps using ArcGIS Pro.	
Complete City Project (Kota Lengkap)	Mar 2024 - Apr 2024
Collaboration Project with BPN Surabaya II and Department of Geomatics Engineering, ITS	
- Conducted cadastral data cleaning and validation in Gading Village, Surabaya. - Identified 121 topological anomalies (104 overlaps and 17 gaps) and performed parcel geometry corrections. - Synchronized ArcGIS shapefiles with AutoCAD working drawings.	

ADDITIONAL INFORMATION

Technical Skills: ArcGIS Pro, QGIS, ENVI, Agisoft Metashape, Autodesk Civil 3D, Google Earth Pro, Leica Infinity, HYPACK, Surfer, MATLAB, Python (Basic), Visual Studio Code, HTML, CSS, JavaScript, Microsoft Office, and Vercel Deployment. Languages: Bahasa Indonesia (Native), English (Professional Working Proficiency)
--



ABDULLAH RACHMAT SUGI

Address : Surabaya, East Java, Indonesia
Phone : +62 858 5359 9295
Email : rachmatsugi09@gmail.com
Portfolio : <https://its.id/m/rachmatsugi>

SUMMARY

Fresh Graduate in Geomatics Engineering from Institut Teknologi Sepuluh Nopember (ITS) with a GPA of 3.41/4.00 and a Very Satisfactory distinction, with experience in GIS, surveying and mapping, and geospatial data management and analysis. Experienced through internships at ATR/BPN and PT Kereta Api Indonesia in GNSS surveying, land asset mapping and verification, cadastral data validation, and spatial data processing using ArcGIS and AutoCAD. Additional project experience includes remote sensing, UAV photogrammetry, and 3D modeling, with the ability to process, analyze, and validate geospatial data to produce accurate spatial information and support mapping and analytical needs.

EDUCATION

Sepuluh Nopember Institute of Technology	Jul 2022 - Aug 2026
Bachelor of Engineering in Geomatics Engineering	
- GPA: 3.41/4.00 - Thesis on "3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method"	
SMA Negeri 1 Bangsal	Jul 2019 - May 2026
Science Major (MIPA)	
Average Grade : 90.23/100	

INTERNSHIP EXPERIENCE

PT Kereta Api Indonesia (Persero) Regional Operation VII Madiun	Jul 2025 - Aug 2025
- Conducted terrestrial boundary surveys for PT KAI land assets on approximately 30 parcels in the Jombang Station area and Jl. Anggrek Madiun using EFIX GNSS RTK for field boundary verification. - Processed survey data and spatial digitization using AutoCAD Civil 3D and ArcGIS Pro to delineate asset boundaries, integrate administrative data, and analyze corridor areas for operational and non-operational lines. - Developed 6 thematic maps and 1 survey layout depicting station distributions, land asset boundaries, and railway corridors across Daop 7 Madiun.	
National Land Agency (ATR/BPN) of Kediri City	Jan 2025 - Feb 2025
- Processed and updated cadastral maps using AutoCAD and ArcGIS to support land registration data maintenance across Kediri City. - Conducted terrestrial field boundary surveys using GNSS for residential housing areas and built structures in Kediri City for on-site boundary verification. - Cross-checked spatial parcel data against land certificates (SHM) to ensure consistency between physical boundaries and legal ownership documents.	
National Land Agency (ATR/BPN) of Batu City	Jul 2024 - Aug 2024
- Performed topological validation and data cleaning on 115 land parcels to eliminate gaps and overlaps, ensuring spatial accuracy and database consistency. - Conducted field boundary surveys using geodetic GNSS equipment across 10 plantation parcels in Sumberbrantas Village, Bumiaji District. - Validated land ownership records by cross-referencing freehold certificates (Sertifikat Hak Milik/SHM) with the institutional cadastral database to ensure legal tenure and data integrity.	

ORGANIZATIONAL EXPERIENCE

- Student Legislative Board, Faculty of Civil, Planning, and Geo Engineering
(BLM FT-SPK ITS)

Nov 2024 - Feb 2026
- Served as Session Chairperson and organizing committee member of the Musyawarah Tingkat Tinggi (MTT) of FT-SPK ITS.
 - Managed administrative affairs and official correspondence to support the preparation and implementation of organizational activities.
 - Coordinated with stakeholders to ensure the effective execution of activities and orderly decision-making processes in accordance with organizational procedures.

- Student Representative Council, Geomatics Engineering Student Association
(BPM HIMAGE-ITS)

Dec 2023 - Dec 2025
- Developed the General Guidelines of Organizational Direction (GBHK) for the HIMAGE-ITS management period.
 - Reviewed and formulated organizational policies and regulations to support effective governance.
 - Guided commission members and coordinated legislative functions in policy formulation and organizational decision-making.

CERTIFICATIONS & TRAINING

- Small Unmanned Aircraft System Remote Pilot Course

PT Terra Drone Indonesia | 2026
- Drone Basic Knowledge & Drone Training Mapping

PVIA ITS | 2023–2024
- Satellite Imagery Data Processing with ENVI

HIMAGE-ITS | 2024

PROJECTS

- 3D Modeling of Candi Tikus Using Integrated UAV and Terrestrial Photogrammetry Data Based on the Gaussian Splatting Method

Nov 2025 - June 2026
- Bachelor's Thesis Project

 - Integrated 1,543 UAV and terrestrial photogrammetry images using 3D Gaussian Splatting (3DGS) across 20,000 optimization iterations.
 - Produced interactive 3D models with 70% smaller file sizes (1.03 GB vs 3.47 GB) and 4–6x faster rendering speeds (92–98 fps vs 16–50 fps) compared to SfM/MVS
 - Achieved centimeter-level geometric accuracy with an RMSE of 1.36 cm and a Cloud-to-Cloud mean distance of 0.183 m, validated across 7 field measurement segments.
- Complete City Project (Kota Lengkap)

Mar 2024 - Apr 2024
- Collaboration Project with BPN Surabaya II and Department of Geomatics Engineering, ITS

 - Participated in the Complete City (Kota Lengkap) cadastral data validation program with ATR/BPN Surabaya II in Gading, Tambaksari District.
 - Identified and analyzed 121 spatial topology errors, consisting of 104 overlap cases and 17 gap anomalies across cadastral parcel boundaries.
 - Executed spatial corrections through boundary repositioning, duplicate/invalid NIB elimination, and cross-synchronized ArcGIS shapefiles with AutoCAD base drawings to ensure geodatabase accuracy.

ADDITIONAL INFORMATION

- **Technical Skills:** ArcGIS Pro, QGIS, ENVI, Agisoft Metashape, Autodesk Civil 3D, Google Earth Pro, Leica Infinity, HYPACK, Surfer, MATLAB, Python (Basic), Visual Studio Code, HTML, CSS, JavaScript, Microsoft Office, and Vercel Deployment.
- **Languages:** Bahasa Indonesia (Native), English (Professional Working Proficiency)